Machine Learning Interatomic Potential for NiCoCr-based alloys

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I. Introduction

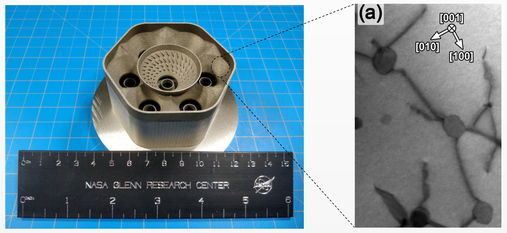


Figure 1 – ODS NiCoCr alloys developed by NASA[1] and STEM image of Y_2O_3 dispersoids in GRX-810[2]

- Machine Learning Interatomic potentials (MLIPs) are data driven method achieving accuracy comparable to DFT while it is much less expensive.
- Universal-MLIPs (U-MLIPs) are MLIPs pretrained on large and diverse datasets so they can cover across many atomic compositions and elements.
- NiCoCr-based alloys are attractive since they have shown incredible creep resistance and high temperature mechanical performance[3].
- However, their chemically complicated structures make them challenging for MLIPs to model accurately.

Research Gap: Although U-MLIPs can cover diverse structures, they still struggle to learn large and complex atomic systems with multiple interactions.

Goal: build model that can capture behavior of NiCoCr-based alloys accurately .

II. Methods

- We selected NEP version 4 model[4], and pretrained it on two open-source major Potential Energy Surface(PES) datasets, MatPES and MP-ALOE.
- For the optimization method of NEP model, we used gradient based optimization instead of SNES(separable neural evolutional strategy) for a faster convergence of training using TorchNEP[5].
- After that, fine-tune the model with the dataset we generate including yttria slabs and various defects structures
- Composition of NiCoCr based alloys used in this study
  Major elements: Ni, Co, Cr
  Minor elements: W, Re, Nb, Ti, Al, O, Yi

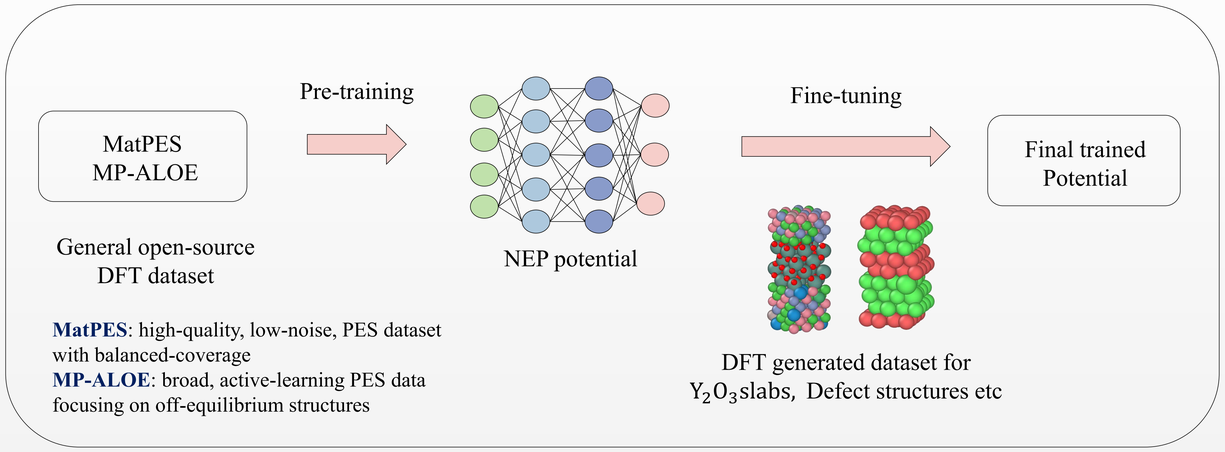


Figure 2 – Workflow of pre-training and fine-tuning of NEP model

III. Model

NEP (Neuroevolutional Potential) model is one of the most efficient MLIPs, which has unique architecture and optimization method.

Architecture: They have fixed non-trainable functions in descriptor layer where the model capture geometric relationship from each atom to neighbors followed by 1 layer feedforward neural network, which allows NEP potential to do computation really fast.

Optimization method: NEP model usually use SNES optimization method, which is derivative free and population-based black-box optimization, which is slow. TorchNEP is using same NEP architecture, but replace SNES to analytical gradient-based optimization, which is faster.

$\theta^{(1)},\theta^{(2)},\ldots,\theta^{(n)} \overset{\text{iid}}{\sim} \mathcal{N}(\mu,\mathrm{diag}(\sigma^2))$
Rank the samples based on loss, and update μ, sigma

SNES

$\nabla_\theta L(\theta)$: Gradient of Loss respect to parameter θ
$\theta \longleftarrow \theta - \eta\nabla\theta L(\theta)$
Update θ

Gradient based

IV. Results

- DFT computation for fine-tuning is now under process.
- Pre-training of NEP model on MATPES and MP-ALOE went well, and converged within 300 epochs.
- It showed good results in parity plots in figure 4, especially trained model has less error in forces than NEP potential with SNES.
- I ran comparison test of change in lattice constant of bulk NiCoCr-based alloy between NEP89 model and out trained model and Our model showed more valid thermal expansion.

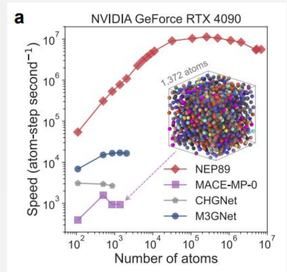


Figure 3 – Time efficiency of NEP89 in MD[6]

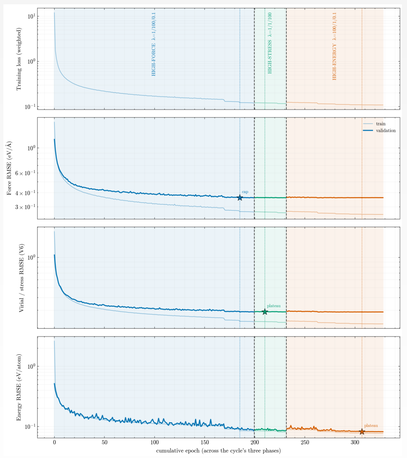


Figure 4 – Training history of NEP model with torch NEP

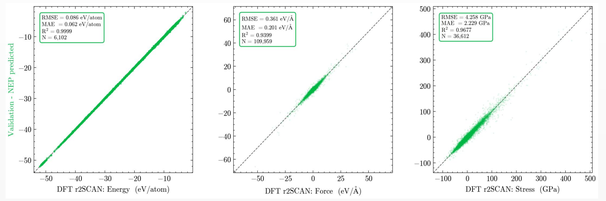


Figure 5 – Parity plots between DFT and trained NEP in energy, force, and stress

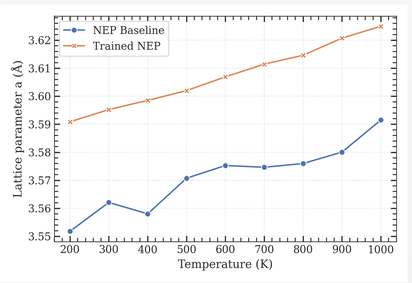


Figure 6 – Comparison of change in lattice constant over temperature between NEP89 base model and trained model

V. Conclusions & Next steps

- Once DFT calculation for fine-tuning dataset is done, we will train NEP model, and do validation.
- Since DFT calculations takes time, and limited to small system with around 100 atoms, we need to carefully select structures that covers diversity, and have broad representation.
- Therefore, we need to figure out the method to sample representative small pieces of structures for one's interesting structures. This will be the gap we would like to fill in next step.

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